

Phase 2 Proposal Document

Project Domain: Agriculture and Plant Disease Detection

Project Number: 10

Project Title: Livestock Skin Disease Image Classifier

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1. Background and Motivation

Livestock health is a critical component of modern agriculture. Skin diseases in livestock animals can significantly affect productivity, animal welfare, and economic outcomes for farmers. Early detection and diagnosis of skin diseases are essential for timely treatment and disease control. Traditional diagnosis methods often require expert veterinary knowledge and can be time-consuming.

Recent developments in Artificial Intelligence (AI) and Deep Learning have enabled automated image-based disease detection systems. Convolutional Neural Networks (CNNs) have shown excellent performance in image classification tasks and can assist in identifying disease patterns from animal skin images. This project aims to develop an intelligent skin disease classification system using deep learning techniques.

2. Problem Statement

Manual identification of livestock skin diseases can be difficult, time-consuming, and dependent on expert veterinary knowledge. There is a need for an automated classification system capable of detecting skin diseases from images with high accuracy. This project proposes the use of CNN and MobileNetV2 transfer learning techniques to classify skin disease images automatically.

3. Selected Dataset and Dataset Link

Dataset Used

Dog Skin Diseases Dataset (used as a veterinary skin disease image dataset for implementing livestock skin disease classification techniques)

Dataset Source

Kaggle

Dataset Link: <https://www.kaggle.com/datasets/youssefmohammed/dogs-skin-diseases-image-dataset>

4. Dataset Description

The dataset contains images of different animal skin disease categories and healthy skin conditions.

Classes

- 1. Dermatitis**
- 2. Fungal Infections**
- 3. Healthy**
- 4. Hypersensitivity**
- 5. Demodicosis**
- 6. Ringworm**

Dataset Organization

The dataset is divided into:

- **Training Dataset**
- **Validation Dataset**
- **Testing Dataset**

Image Format

- **JPG Images**
- **RGB Color Images**

Image Size

All images are resized to:

224 × 224 pixels

for model training.

5. Research Questions

- 1. Can deep learning models accurately classify skin diseases from animal skin images?**
- 2. Does transfer learning improve disease classification performance?**
- 3. Which model performs better: Custom CNN or MobileNetV2?**
- 4. What evaluation metrics best measure classification effectiveness?**

6. Expected Results

The project is expected to:

- **Successfully classify multiple skin disease categories.**
- **Achieve high classification accuracy.**
- **Demonstrate the effectiveness of transfer learning.**

- Generate confusion matrices and classification reports for performance evaluation.
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7. Proposed Methodology

Step 1: Dataset Collection

Obtain the skin disease image dataset from Kaggle.

Step 2: Data Preprocessing

- Image resizing
- Image normalization
- Data cleaning

Step 3: Data Augmentation

Apply:

- Rotation
- Zooming
- Horizontal Flipping

to improve model generalization.

Step 4: Dataset Splitting

Use:

- Training Dataset
- Validation Dataset
- Testing Dataset

Step 5: CNN Development

Develop a custom Convolutional Neural Network for disease classification.

Step 6: Transfer Learning

Implement MobileNetV2 using pretrained ImageNet weights.

Step 7: Model Training

Train both models using TensorFlow and Keras.

Step 8: Model Evaluation

Evaluate performance using multiple metrics.

8. CNN Model Design

CNN Architecture

Input Layer ($224 \times 224 \times 3$)



Convolution Layer (32 Filters)



Max Pooling



Convolution Layer (64 Filters)



Max Pooling



Convolution Layer (128 Filters)



Max Pooling



Flatten Layer



Dense Layer



Dropout Layer



Output Layer (6 Classes)

9. Transfer Learning Model

MobileNetV2

The MobileNetV2 model uses pretrained ImageNet weights for feature extraction.

Configuration:

- **MobileNetV2 Base Model**
- **Global Average Pooling Layer**
- **Dense Layer (128 Units)**
- **Softmax Output Layer (6 Classes)**

Benefits:

- **Faster convergence**
 - **Better feature extraction**
 - **Improved classification accuracy**
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10. Evaluation Metrics

The following metrics will be used:

Accuracy

Measures overall prediction correctness.

Precision

Measures the correctness of positive predictions.

Recall

Measures the ability to identify actual positive samples.

F1-Score

Balances Precision and Recall.

Confusion Matrix

Provides detailed class-wise performance evaluation.

11. Expected Figures

- 1. Sample Dataset Images**
 - 2. CNN Architecture Diagram**
 - 3. MobileNetV2 Architecture Diagram**
 - 4. Accuracy Curve**
 - 5. Loss Curve**
 - 6. Confusion Matrix**
 - 7. Model Comparison Graph**
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12. Expected Tables

- 1. Dataset Summary Table**
- 2. CNN Performance Metrics Table**
- 3. MobileNetV2 Performance Metrics Table**
- 4. CNN vs MobileNetV2 Comparison Table**

Tools and Technologies

- **Python**
- **TensorFlow**
- **Keras**
- **NumPy**
- **Pandas**
- **Matplotlib**
- **Seaborn**
- **Scikit-Learn**
- **Kaggle Notebook**

Expected Outcome

The proposed system will automatically classify livestock-related skin disease images using deep learning techniques. The comparison between CNN and MobileNetV2 will demonstrate the effectiveness of transfer learning in disease classification tasks and support intelligent veterinary diagnostic systems.